

**Linear Graphs –
Plotting $y = mx + c$
The Magic Rule of
Straight Lines**



Meet Rohan's Auto Ride!

Rohan takes an auto to the cricket ground.

How do we find the fare for his ride?



Hint: Fare = Base fare + (Rate $\times$ Distance)

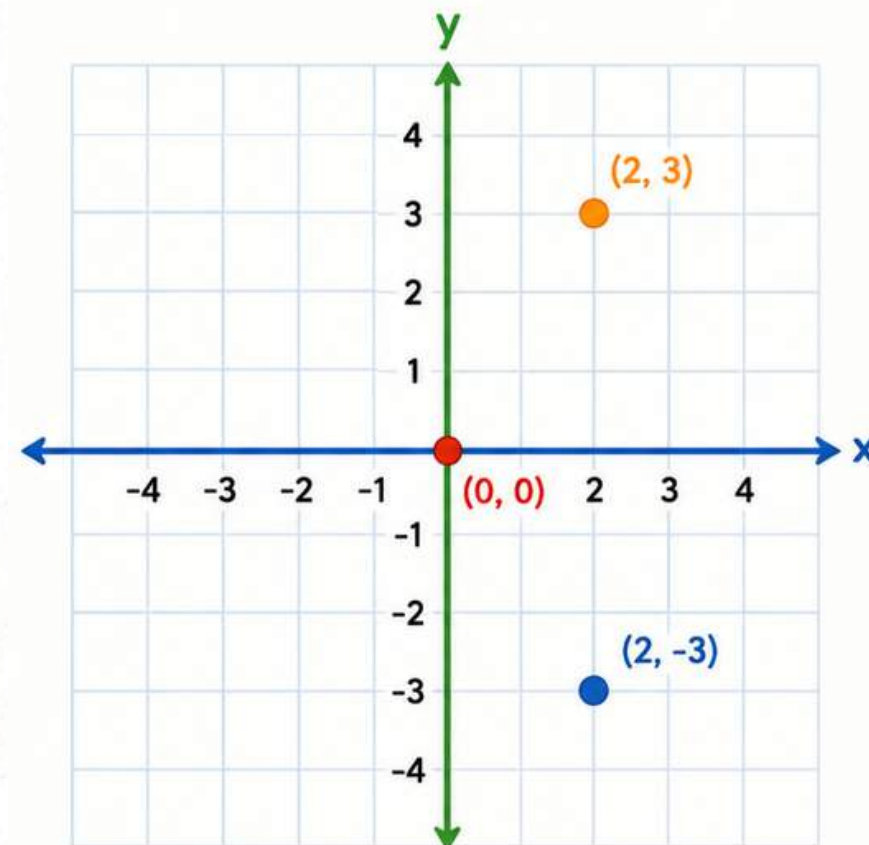


The Magic Map – Coordinate Plane

A map made of two lines:

- x-axis (left to right)
- y-axis (bottom to top)

Where they meet is called the Origin (0, 0)!



Like a treasure map
showing where to find
points!

Every point has an
address: (x, y)

What is $y = mx + c$?

This is the magic rule for straight lines!



x = number we choose (input)



y = answer we get (output)



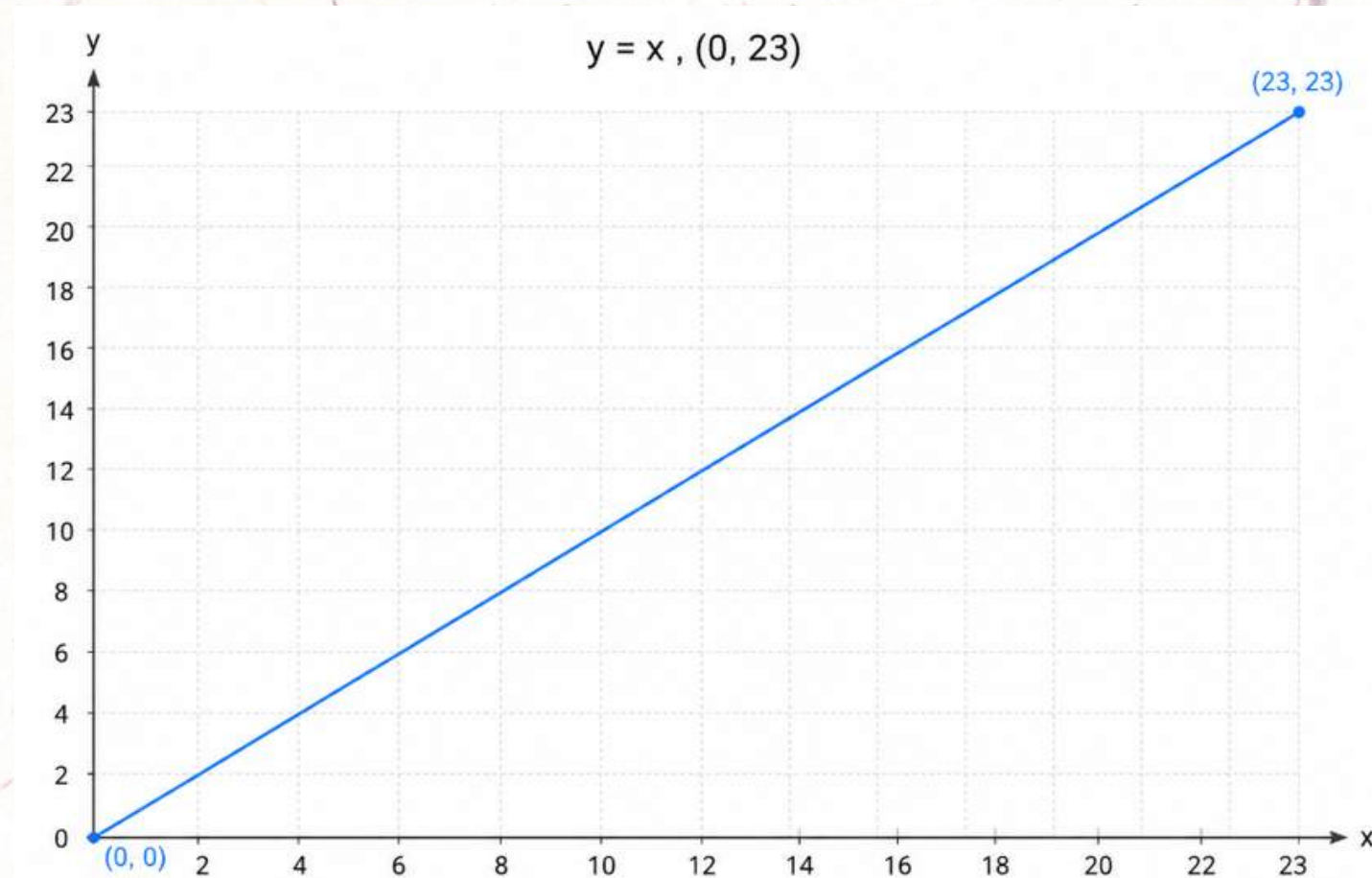
m = steepness (how fast y grows)



c = starting value (where line begins)

$$y = mx + c$$

Understanding 'Starting Value' c



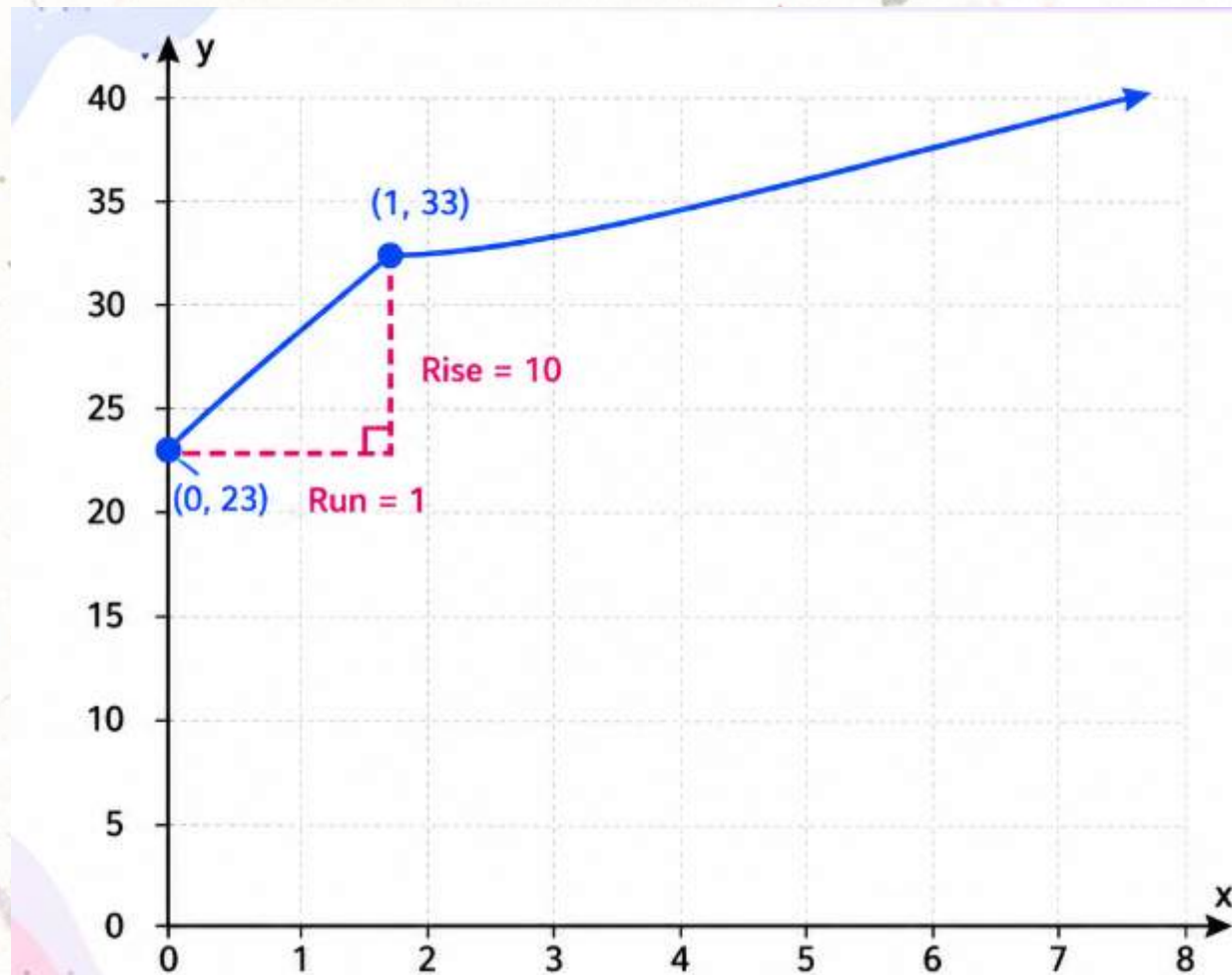
c is where the line starts when $x = 0$.

Auto base fare = ₹23 when distance is 0 km.

Even before travelling 1 km, the meter starts at ₹23. That is c — the starting value on the y-axis!



Understanding 'Steepness' m



m = how much y goes up when x goes up by 1

Auto ride: ₹10 per extra km!
When x goes from $0 \rightarrow 1$, y goes
from $23 \rightarrow 33$.
Each step up = ₹10 more fare.



Let's Calculate $y = 2x + 1$!

Let's find y when $x = 0, 1, 2$.
Substitute each x into $y = 2x + 1$

$$x = 0 \rightarrow y = 2(0) + 1 = 1$$

$$x = 1 \rightarrow y = 2(1) + 1 = 3$$

$$x = 2 \rightarrow y = 2(2) + 1 = 5$$

$$y = 2x + 1$$

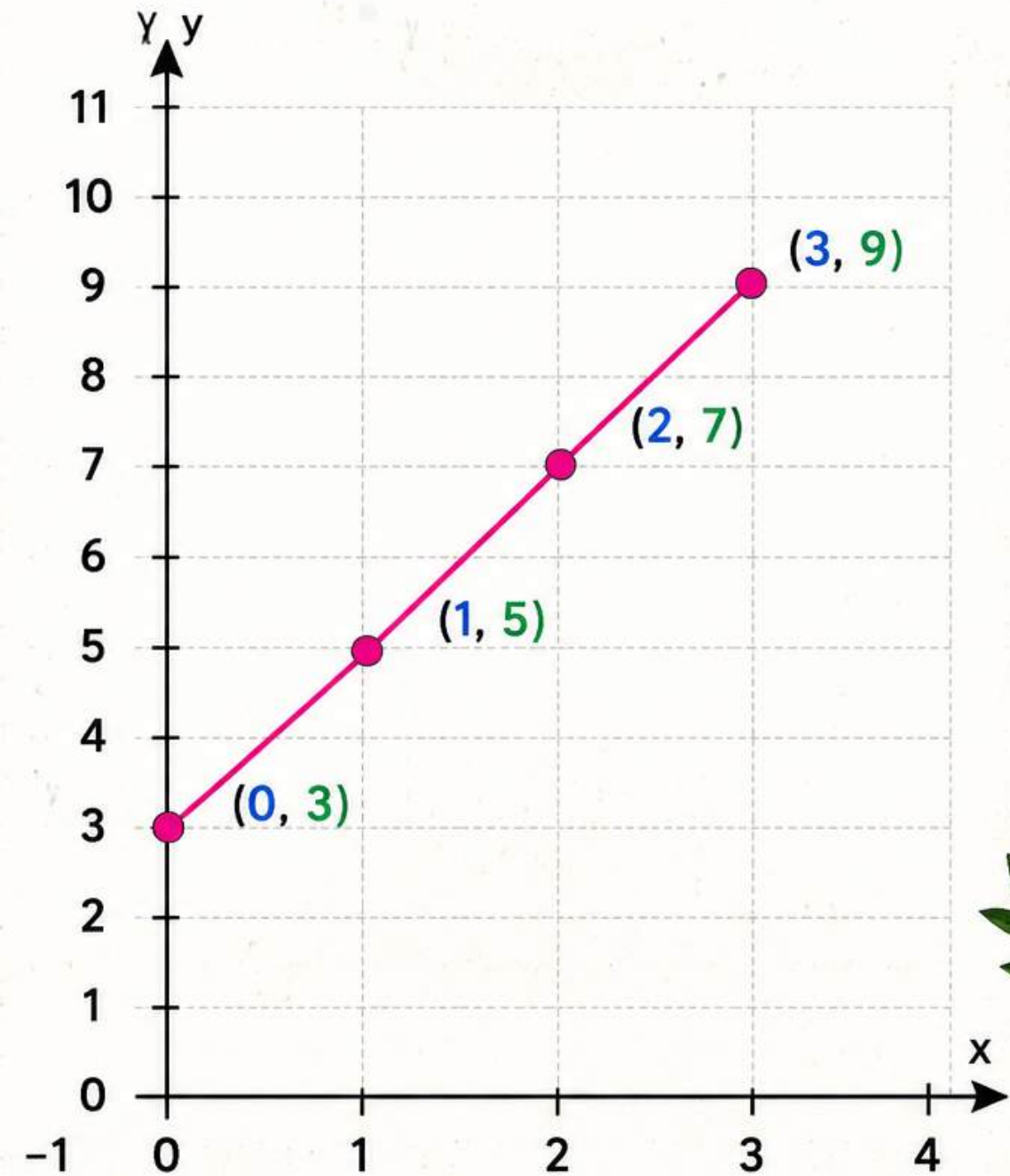
x	$y = 2x + 1$
0	1
1	3
2	5



Plotting Points – Paper Flowers!

Plot these points on the grid and connect them!
 $y = 2x + 3$

$(0,3)$ $(1,5)$ $(2,7)$ $(3,9)$



Real-Life Example – Auto Fare!



Fare = ₹43

$$\text{Auto fare} = 10 \times \text{km} + 23$$

If Rohan travels 2 km:

$$y = 10 \times 2 + 23$$

$$y = 20 + 23$$

$$= ₹43 \text{ 🎉}$$

Interactive Challenge – Predict the Y-value!

If $y = 3x + 2$, what is y when $x = 2$?
Tap the correct answer below!

A) 6 B) 8 C) 10 D) 12

Find y when $x = 2$

$$y = 3x + 2$$

$$y = 3(2) + 2$$

$$y = 6 + 2$$

$$Y = ?$$





Quick Poll – Your Turn!

Option A

x
(the input number)

Option B

m
(the steepness)

Option C ✓

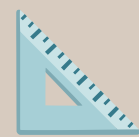
c
(the starting value!)

Recap – The Magic Rule of Straight Lines ✨

$y = mx + c$ means:



x: input number



m: steepness or step size



c: starting value (where line begins)



Plot points → connect → straight line!

Thank You!
Any Questions?

